

Practical 11

Postlab (OPEN ENDED PRACTICAL)

Aim: Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

CODE:-

```
chanchal-manwatkar@chanchal-manwatkar-  
GNU nano 7.2 postlab.c *  
#include <stdio.h>  
  
#define MAX 50  
  
int main() {  
    int pages[MAX], frames[MAX];  
    int n, f, i, j, k = 0, pageFaults = 0;  
    int found;  
  
    // Input number of pages  
    printf("Enter number of pages: ");  
    scanf("%d", &n);  
  
    // Input page reference string  
    printf("Enter page reference string:\n");  
    for(i = 0; i < n; i++) {  
        scanf("%d", &pages[i]);  
    }  
  
    // Input number of frames  
    printf("Enter number of frames: ");  
    scanf("%d", &f);  
  
    // Initialize frames with -1 (empty)  
    for(i = 0; i < f; i++) {  
        frames[i] = -1;  
    }  
  
    printf("\nPage\tFrames\n");  
  
    // Demand Paging with FIFO  
    for(i = 0; i < n; i++) {  
        found = 0;
```

```
chanchal-manwatkar@chanchal-manwatkar-
GNU nano 7.2 postlab.c *
    found = 0;

    // Check if page already in frame
    for(j = 0; j < f; j++) {
        if(frames[j] == pages[i]) {
            found = 1;
            break;
        }
    }

    // If not found → Page Fault
    if(!found) {
        frames[k] = pages[i]; // Load page into frame
        k = (k + 1) % f;      // FIFO replacement
        pageFaults++;
    }

    // Display frames
    printf("%d\t", pages[i]);
    for(j = 0; j < f; j++) {
        if(frames[j] != -1)
            printf("%d ", frames[j]);
        else
            printf("- ");
    }
    printf("\n");
}

printf("\nTotal Page Faults = %d\n", pageFaults);

return 0;
}
```

OUTPUT:-

```
chanchal-manwatkar@chanchal-manwatkar-Aspire-A515-56G:~$ nano postlab.c
chanchal-manwatkar@chanchal-manwatkar-Aspire-A515-56G:~$ gcc postlab.c -o postlab.c
gcc: fatal error: input file 'postlab.c' is the same as output file
compilation terminated.
chanchal-manwatkar@chanchal-manwatkar-Aspire-A515-56G:~$ gcc postlab.c -o postlab
chanchal-manwatkar@chanchal-manwatkar-Aspire-A515-56G:~$ ./postlab.c
bash: ./postlab.c: Permission denied
chanchal-manwatkar@chanchal-manwatkar-Aspire-A515-56G:~$ ./postlab
Enter number of pages: 6
Enter page reference string:
1 2 3 1 4 5
Enter number of frames: 4

Page   Frames
1      1 - - -
2      1 2 - -
3      1 2 3 -
4      1 2 3 -
5      1 2 3 4
6      5 2 3 4

Total Page Faults = 5
```